INCH-POUND

A-A-52213
January 10, 1997

COMMERCIAL ITEM DESCRIPTION

BATH UNIT, PORTABLE, FOUR SHOWERHEADS

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers portable, fuel-fired showers (4) contained in a shelter intended to provide hot showers to soldiers in the field for hygienic purposes.

2. SALIENT CHARACTERISTICS

2.1 Design. The Portable Bath Unit (PBU) shall consist of a shelter which contains four shower stalls and a separate water heating unit. The PBU, when configured for transportation, shall consist of no more than three separate modules, each weighing 75 pounds or less. Handles or other suitable means for lifting and carrying shall be provided on each module. Overall dimensions shall be such that each module is able to be carried by two people. The PBU shall be transportable by High Mobility Multipurpose Wheeled Vehicle (HMMWV), Commercial Utility Cargo Vehicle (CUCV) and larger military vehicles. The PBU shall be capable of being set-up and made ready for operation by two experienced people in a maximum time of 15 minutes (see 4.3.2). The PBU shall also be capable of being dismantled by two experienced people in a maximum time of 15 minutes (see 4.3.2). The PBU shall be capable of starting and operating in ambient temperatures of 32°F to 120°F. No special tools shall be required for set-up or operation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, ATTN: SSCNC-WEF, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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All component parts shall be interchangeable with the same parts of others units furnished on the same contract. The PBU shall incorporate safety features which are adequate to avoid hazardous operation.

2.2 Heater. The heater shall draw, heat, and dispense water to the shower in accordance with 2.2.1 and 2.3, and shall be capable of pumping water without the heater in operation. It shall burn diesel and JP-8 fuels for water heating. Electrical components shall operate on nominal 24/28V DC with a maximum draw of 15 amps power. Burner exhaust shall be vented away from the heater module controls, and away from the shower area, such that exhaust gases do not create a health hazard during operation. Fuel and water connections shall be a self-sealing, quick-disconnect type in accordance with the manufacturers standard practice. Means for water temperature adjustment and control, and an outlet water temperature gauge shall be provided. Means shall be provided to prevent excessive pressure build up inside the heater module. The heater module shall be a single self-contained unit with maximum dimensions 36 inches long, 18 inches wide, and 22 inches high. Sound level shall be measured one foot from the center of the top surface of the heater and shall not exceed 85 dB(A).

2.2.1 Water supply. The heater shall be capable of drawing water from the standard 5-gallon military water can (NSN 7240-00-089-3827), from surface water sources up to 10 feet away and 5 feet below the inlet connection, and from pressurized sources such as tanks and municipal taps. Adapters shall be provided for connection to the M149A2 400 gallon water tank trailer (NSN 2330-01-108-7367) equipped with a standard 1 inch nominal male garden hose connector, with thread type NH conforming to ASME B1.20.7; and the 160 gallon collapsible pillow tank equipped with a 2 inch nominal cam-locking quick disconnect male coupling. An inlet water strainer to remove solid particles, which is removable for cleaning, shall also be provided.

2.2.2 Fuel system. The PBU shall operate on both diesel and JP-8 fuels. Fuel consumption rate shall not exceed ½ gallon per hour. The heater shall be capable of drawing fuel from the standard military 5-gallon plastic fuel can (NSN 7240-01-337-5269, 7240-01-337-5268). An adapter cap which interfaces with the can, and provides means for drawing fuel from the can when in an upright position shall be provided (see 6.4). Hoses with metal quick-disconnect fittings shall be provided to connect the heater module with the adapter. All components shall be resistant to deformation or deterioration from exposure to fuels. A replaceable fuel filter shall be provided.

2.2.3 Over temperature protection. Water temperature in any part of the system shall not exceed 200°F.

2.2.4 Electrical supply ("Government unique" requirement, see 6.3). The heater module shall be capable of drawing electrical power from military vehicles via connection to the NATO standard slave receptacle (see 6.4). A detachable, insulated, waterproof, power cable, minimum length of 10 feet, with a connector for mating with the vehicle receptacle shall be provided.

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2.2.5 Internal wiring. Wiring practices shall be in accordance with NFPA Standard No. 70.

2.2.6 Power switch. A main power switch which interrupts power to all internal components when switched off shall be provided. A green indicator light shall be provided for this switch indicating that the system is energized.

2.2.7 Circuit protection. The electrical system shall include replaceable fuses or circuit breakers which prevent damage to all electrical components and injury to personnel in the event of current surges.

2.2.8 Self-diagnostics. Self-diagnostics indicating, differentiating, and identifying system and component faults shall be incorporated in the electrical system. At a minimum, the following faults shall be identified: failure to start, loss of burner flame, water over temperature, and low voltage.

2.3 Operation of shower. The heater shall provide water to the shower heads at a temperature of between 95°F and 105°F within 15 minutes. The heater shall continuously maintain this temperature range over the full range of flow conditions and inlet water temperatures from 32°F to 95°F (see 4.3.1). The outlet water temperature shall be adjustable at the heater module.

2.4 Accessory storage. A bag or a transit case for storage of all associated hoses, cables, and accessories during transport shall be provided. The weight of the accessories and container shall be included with the weight of the heater.

2.5 Shelter. The shelter module shall consist of a lightweight, tent-type structure with a floor. Maximum external dimensions when erected shall be 12 feet by 12 feet. Staking and tie down provisions shall be provided with each shelter. The entrance of the shelter, when erected, shall be a minimum of 75 inches high. There shall be a minimum of 75 inches in height from the floor to the roof over the shower stalls. Means for temporary dry storage of clothing (excluding helmet) during showering shall be provided. The exterior color of the shelter shall be as specified in the contract or purchase order (see 6.1). The shelter, when configured for transport, shall form a single package with maximum dimensions of 48 inches long by 44 inches wide by 24 inches high. A bag or a transit case shall be provided for transport. The shelter and shower may be transported as a single package provided it does not exceed the above dimensions and a combined weight of 75 pounds (see 2.1).

2.5.1 Fenestration. A window with a cover flap shall be provided in the wall and roof in the center of each individual shower area. Zippered openings shall be provided in each wall which allow walls to be flapped open for ventilation and for entry/exit. Screened vents with covers shall be provided in walls and roof near the top of each individual shower area such that each stall has ventilation openings with a minimum total area of one sq. ft. each.

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2.6 Showers. The four showers shall each consist of a water distribution hose, shower head, waste water drainage apparatus, partition curtains, supporting structure, and all associated hardware. The showers shall be detachable from the shelter, and be capable of operating without the shelter.

2.6.1 Shower heads. Four hand-held shower heads shall be provided. Each head shall be equipped with a push-button "on/off flow" control which remains in the normal off condition unless activated by the user.

2.6.2 Partitions. Partitions shall be provided for privacy and containment of water within the individual shower area. Means for holding soap, toiletries, etc. during showering shall be provided in each stall. Means for mounting both a military standard lightweight flashlight or a chemlight for showering at night shall be provided in each stall at a height of 60-70 inches.

2.6.3 Flooring. The shower floor shall be designed to prevent the accumulation of waste water in individual showers. General purpose floor mats conforming to NSF International Standard No. 52 shall be provided in shower areas and walkways and shall be easily removable for cleaning.

2.6.4 Drainage. Means for segregating, collecting, and draining of used water from individual shower areas and from the shelter shall be provided. The system shall provide a single outlet point.

2.7 Marking.

2.7.1 Identification plate ("Government unique" requirement, see 6.3). The heater module shall have a metal identification plate permanently attached to the outside of the unit enclosure. The material shall be aluminum base alloy of .020 inches thickness. The identification plate shall contain the following information:

1. Serial number.
2. Contract number.
3. Nomenclature (item name and type designation).
4. Manufacturer's name and address.
5. Model number.
6. CAGE number (manufacturer's identification).
7. National Stock Number (NSN).
8. Government ownership designation (U.S.)

Lettering shall be upper case, a minimum of 1/8 inch in height, and shall be permanently inscribed.

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Each plate shall be placed to be readily visible to the operator during normal operation, and shall not adversely affect the life of the PBU.

2.7.2 Instruction plates. Instruction plates shall be permanently attached to the water heating unit. The instruction plates shall include the operating instructions and the following warnings:

CAUTION: Use potable water only.

CAUTION: Use diesel or JP-8 fuels only.

CAUTION: Do not operate in enclosed areas. Direct exhaust away from shelter.

CAUTION: Parts of heater module may become hot during operation.

CAUTION: Always flush, drain and clean heater and hoses completely following use.

CAUTION: Evacuate and lower shelter if wind conditions exceed 30 mph.

2.7.3 Labels. Controls shall be visible and permanently labeled. Lettering shall be upper case, and not less than 1/8 inch in height. Each module as configured for transport shall be visibly and permanently labeled with its weight and the number of personnel required for lifting using 1/4 inch high uppercase lettering. Safety labels which meet ANSI Z535.4 shall be used.

2.8 Operator's manual ("Government unique" requirement, see 6.3). An operator's manual which clearly describes system set-up, operation, cleaning and minor maintenance shall be provided. The operator's manual shall not require reading skills higher than tenth grade level and shall employ illustrations to ensure clarity. The operator's manual shall include a warning to use only Army Medical Department (AMEDD) approved potable water and a warning to operate the PBU in well-ventilated, outdoor areas only. The operator's manual shall also include the following statement in the cleaning section: "Clean showers using general detergent and hot water. Scrub all surfaces then rinse with hot water. Disinfect shower walls, floors, ceiling and floor mats using a solution of 2 ounces of household bleach in 4 gallons of water. All surfaces must be thoroughly wet with bleach solution and then air dried. (WEAR PROTECTIVE EQUIPMENT)."

2.9 Cleaning. Procedures for cleaning and sanitizing the PBU, including descaling of the heater unit by using readily available cleaning supplies shall be detailed in the operator's manual (see 2.8). Procedures shall cover the heater, shelter and shower modules and all associated hoses. All necessary cleaning equipment, including personal protective equipment such as gloves, if required, shall be provided.

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2.10 Spare parts. Spare parts and repair kits which are normally included commercially shall be provided.

2.11 Materials and components. When materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice provided the completed items comply with all provisions of this document.

2.12 Fabrics. Fabric materials shall be flame, water, and fungus resistant.

2.13 Storage. The PBU shall be capable of withstanding storage temperatures of 160°F for 8 hours and -28°F for 8 hours without failing to operate, or causing permanent set or loss of resiliency of gaskets, or discoloring, cracking or bulging of the unit (see 4.3.1).

2.14 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS

3.1 Contractor recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, except for any "Government unique" requirements, shall conform to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability criteria. No PBU offered under this CID shall be acceptable unless the manufacturer has manufactured the basic unit for a period of one year and sold it to the government or commercially. Minor modification or modifications made to meet Government unique requirements are permitted provided the PBUs are available in time to meet the delivery schedule in the solicitation. Questions regarding the classification of a modification as minor may be addressed to the technical activity identified on page 1 of this CID.

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4.3 Performance tests.

4.3.1 Storage test. The PBU shall satisfy the storage requirements for both hot and cold temperatures as specified in 2.13.

4.3.2 Set-up test. The PBU shall satisfy the set-up requirements specified in 2.1.

4.3.3 Tear-down test. The PBU shall satisfy the tear-down requirements specified in 2.1.

4.3.4 Operational test. The PBU shall satisfy the operational requirements specified in 2.3.

5. PACKAGING

5.1 Packaging. Preservation, packing and marking requirements shall be as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this CID.
b. Color required (see 2.5)
c. Packaging requirements (see 5.1).

6.2 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID are listed below; however, competition is not limited to these companies.

U.S Thermal Inc.
9846 Crescent Park Drive
West Chester, Ohio 45069
(513) 777-7763

6.3 "Government unique" requirement. Whenever a "Government unique" requirement is included in a paragraph under "Salient Characteristics", it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

6.4 Commercial sources. The fuel can swivel adapter, Part No. 98-00000280, NATO Stock Number 7240-21-911-8834, from the source below has been found to meet the requirements of 2.2.2:

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Scepter Manufacturing Company, Limited
170 Midwest Road
Scarborough, Ontario, Canada M1P 3A9

Companies listed below are suggested sources for the slave receptacle (see 2.2.4):

Gateway Cable Company
372 North Meyer Road
Bensenville, IL 60106-1948

General Motors of Canada Ltd. Diesel Division
1991 Oxford Street E.
London, Ontario, Canada N6A 4N5

6.5 Non-Government publications.

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 1.20.7 - Hose Coupling Screw Threads

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, Quincy, MA 02269.)

NSF INTERNATIONAL

NSF International Standard No. 52 - Supplemental Flooring

(Application for copies should be addressed to the NSF International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.)

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ASTM STANDARDS

E 380 - Use of the International System of Units (SI) (the Modernized Metric System)

(Application for copies of ASTM standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

ANSI STANDARDS

ANSI Z535.4 - American National Standard for Product Safety Signs and Labels

(Application for copies of ANSI standards should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

MILITARY INTERESTS:

Custodians

Army - GL
Navy - YD1
Air Force - 99

Review Activities

Army - AV, MD1
Navy -MC
Air Force - 03, 84
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - GL

PROJECT 4510-0295

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	A-A-52213	970110

3. DOCUMENT TITLE
Bath Unit, Portable, Four Showerheads

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial (508) 233-5175	(2) AUTOVON 256-5175
c. ADDRESS *(Include Zip Code)* Commander, U.S. Army Natick RD&E Center ATTN: SSCNC-WEF Natick, MA 01760-5018	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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